

Tools for troubleshooting and theory

- I. List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

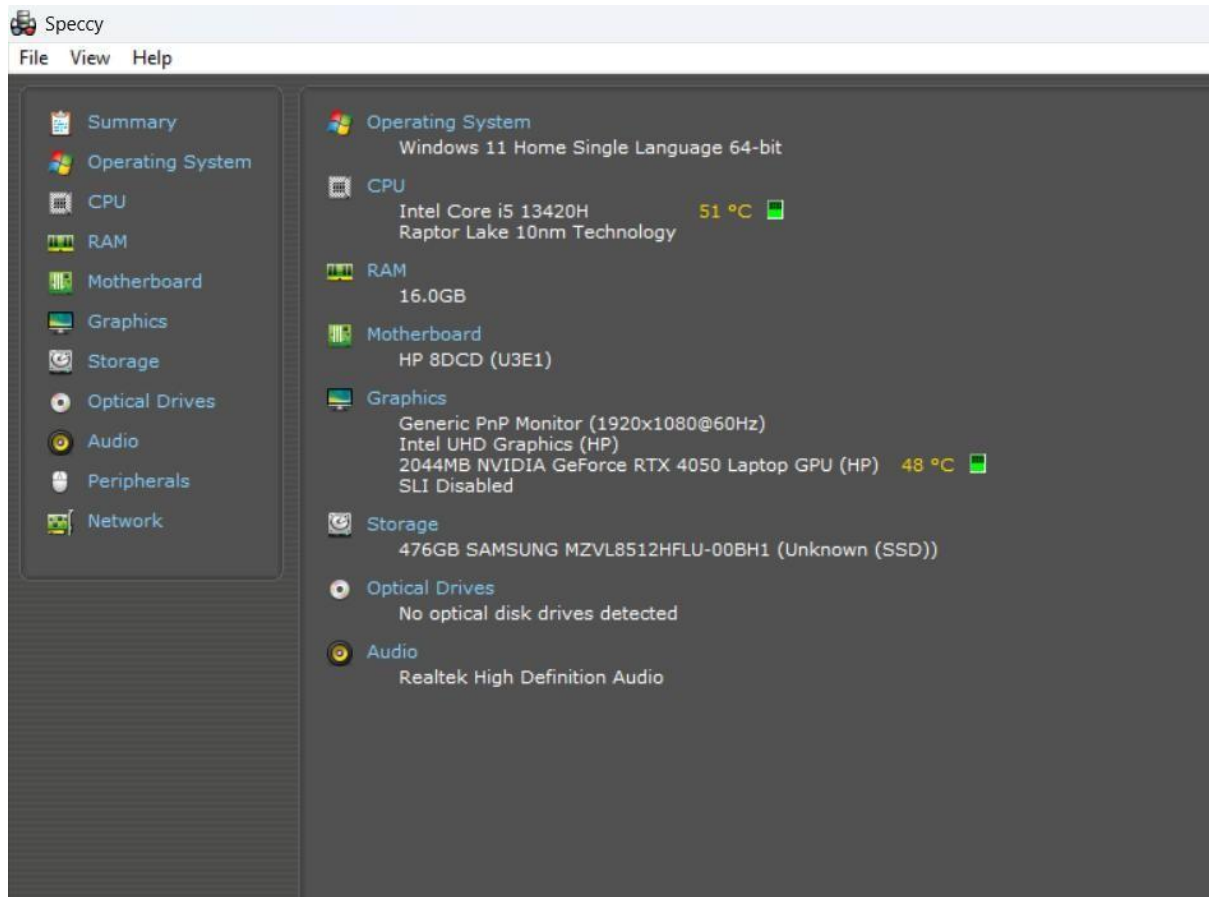
Scenario	How the multimeter helps
Computer not turning on	Checks if the power supply (SMPS) is providing the correct voltage.
Laptop battery not charging	Measures the battery or charger voltage to identify a faulty battery or adapter.
Smartphone not charging	Tests the charger, USB cable, or charging port voltage to find the charging issue.

II. Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video. Hint: Focus on the codes shown and how they relate to hardware issues.

- A POST card is inserted into the motherboard to display POST (Power-On Self-Test) error codes during startup. In the video, the card showed a specific error code that indicated a RAM-related problem. After replacing or reseating the RAM, the system completed the POST successfully and booted normally. The POST card helped identify the faulty hardware quickly.

- III. Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.



- IV. Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Voltage, current, and electrical continuity	Faulty power supply or charger
POST Card	Motherboard POST error codes	RAM or CPU initialization failure
Software Diagnostics	System temperature, storage health, RAM, CPU information	Overheating, failing hard drive, or high CPU temperature